SQL :

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- ✧ **SQL** is a **language** used to manage databases, while **MySQL** is a **software** (a relational database management system) that uses SQL to interact with data.

1. What is SQL?

SQL (Structured Query Language) is a programming language used to manage and manipulate databases. Think of a database as a digital filing cabinet, and SQL as the way to interact with it. You use SQL to:

- Retrieve data (SELECT)
- Insert data (INSERT)
- Update data (UPDATE)
- Delete data (DELETE)

2. Basic Database Concepts

Before diving into SQL commands, it's important to understand some fundamental database terms:

- **Database:** A collection of data organized in tables.
- **Table:** A collection of rows and columns, similar to a spreadsheet.
- **Row (or Record):** A single, horizontal entry in a table (like a student in a class).
- **Column (or Field):** A vertical entry in a table, such as a student's name or age.
- **Primary Key:** A unique identifier for a row in a table (e.g., student ID).
- **Foreign Key:** A column that links one table to another.

1. Data Definition Language (DDL)

DDL commands are used to **define, modify, and manage the structure of database objects** such as tables, indexes, and schemas. These commands are responsible for the **creation, modification, and deletion** of database structures.

Common DDL Commands:

1. **ALTER**: Used to modify an existing database object, such as adding or deleting columns in a table.
   - **Example**:
     ALTER TABLE students ADD COLUMN email VARCHAR(100);
2. **DROP**: Used to delete an existing database object, such as a table, view, or index.
   - **Example**:
   - DROP TABLE students;
3. **TRUNCATE**: Used to remove all rows from a table but **does not remove the table structure** (you can still use the table after truncating it).
   - **Example**:
   - TRUNCATE TABLE students;
4. **RENAME**: Used to rename an existing database object, such as a table or column.
   - **Example**:
     RENAME TABLE students TO students_new;

2. Data Manipulation Language (DML)

DML commands are used to **manipulate the data stored in existing tables**. These commands help you **insert, update, delete, and retrieve** data. Unlike DDL commands, DML commands do not affect the structure of the database objects; they only deal with the data.

Common DML Commands:

1. **SELECT**: Used to retrieve data from one or more tables in the database. It is the most commonly used SQL command.
   - **Example**:

- SELECT * FROM students;

2. **INSERT**: Used to add new rows (records) into a table.
   - **Example**:
   - INSERT INTO students (student_id, name, age, grade, enrolled_date)
     VALUES (1, 'John Doe', 20, 'A', '2023-09-01');

3. **UPDATE**: Used to modify existing records in a table. You can specify conditions to update specific rows.
   - **Example**:
     - UPDATE students
       SET grade = 'B+'
       WHERE student_id = 1;

4. **DELETE**: Used to remove one or more rows from a table based on a condition.
   - **Example**:
   - DELETE FROM students
     WHERE student_id = 1;

3. Basic SQL Commands

SELECT Statement

The SELECT statement is used to query the database and retrieve data.

- **Example:**

  SELECT column1, column2 FROM table_name;
- **Explanation:**
  - SELECT: Tells the database that we want to retrieve data.
  - column1, column2: Specifies the columns we want to see.
  - FROM table_name: Specifies from which table to retrieve the data.
- **Example 1:**

  SELECT name, age FROM students;

INSERT INTO Statement

The INSERT INTO statement is used to add new rows of data to a table.

- **Example:**
- INSERT INTO table_name (column1, column2) VALUES (value1, value2);
- **Explanation:**
  - INSERT INTO: Tells the database to add data.
  - column1, column2: The columns where the values will be inserted.
  - VALUES (value1, value2): The actual data to insert.
- **Example 1:**

  INSERT INTO students (name, age) VALUES ('John Doe', 20);

UPDATE Statement

The UPDATE statement is used to modify existing data in a table.

- **Example:**

  UPDATE table_name SET column1 = value1 WHERE condition;
- **Explanation:**
  - UPDATE: Tells the database to modify data.
  - SET: Specifies which columns to update.
  - WHERE: Specifies the condition to identify which rows to update.
- **Example 1:**

  UPDATE students SET age = 21 WHERE name = 'John Doe';

DELETE Statement
The DELETE statement is used to remove data from a table.

- **Example:**

  DELETE FROM table_name WHERE condition;
- **Explanation:**
  - DELETE FROM: Tells the database to remove data.
  - WHERE: Specifies the condition for which rows to delete.
- **Example 1:**

  DELETE FROM students WHERE name = 'John Doe';

1. **Count how many students are in the table**:

   SELECT COUNT(*) FROM students;

2. **Get the average age of all students**:

   SELECT AVG(age) FROM students;

3. **Get the highest grade (alphabetically)**:

   SELECT MAX(grade) FROM students;

1. **Get all students ordered by age (ascending)**:

   SELECT * FROM students ORDER BY age ASC;

2. **Get all students ordered by age (ascending)**:

   SELECT * FROM students ORDER BY grade DESC;

3. **Get students ordered by grade first, then by age (ascending)**:

   SELECT * FROM students ORDER BY grade DESC, age ASC;

✧ **SAMPLE QUERIES FOR SQL:**

Sample students Table

CREATE TABLE SQL Query

```
CREATE TABLE students (
   student_id INT PRIMARY KEY,
   name VARCHAR(100),
   age INT,
   grade VARCHAR(2),
   enrolled_date DATE
);
```

student_id	name	age	grade	enrolled_date
1	John Doe	20	A	2023-09-01

2	Alice	21	B	2023-09-05
3	Bob	19	C	2023-09-08
4	Charlie	22	B+	2023-09-10
5	Eve	20	A	2023-09-12

✧ ***DDL Examples (Table Structure Level)***

1. ALTER – Add a new column

```
ALTER TABLE students ADD COLUMN email VARCHAR(100);
```

2. DROP – Delete the whole table

```
DROP TABLE students;
```

3. TRUNCATE – Remove all data (keep structure)

```
TRUNCATE TABLE students;
```

4. RENAME – Rename table

```
RENAME TABLE students TO students_backup;
```

DML Examples (Data Handling Level)

1. SELECT – Retrieve all students

```
SELECT * FROM students;
```

2. INSERT – Add a new student

```
INSERT INTO students (student_id, name, age, grade, enrolled_date)
VALUES (6, 'David', 23, 'B', '2023-09-15');
```

3. UPDATE – Change grade of student ID 1

```
UPDATE students
SET grade = 'A+'
WHERE student_id = 1;
```

4. DELETE – Remove student with ID 2

```
DELETE FROM students
WHERE student_id = 2;
```

✧ ***More SELECT Examples (Queries on data)***

1. Get only names and ages

```
SELECT name, age FROM students;
```

2. Count students

```
SELECT COUNT(*) FROM students;
```

-- Output: 5

3. Average age

```
SELECT AVG(age) FROM students;
-- Output: e.g., 20.6
```

4. Maximum grade alphabetically

```
SELECT MAX(grade) FROM students;
-- Output: C
```

5. Students sorted by age ascending

```
SELECT * FROM students ORDER BY age ASC;
```

6. Students sorted by grade descending

```
SELECT * FROM students ORDER BY grade DESC;
```

7. Students sorted by grade DESC, then age ASC

```
SELECT * FROM students ORDER BY grade DESC, age ASC;
```

✧ ***Combined Example Query***

Get students older than 20 with grade 'B' or higher, sorted by enrolled_date

```
SELECT * FROM students
WHERE age > 20 AND grade >= 'B'
ORDER BY enrolled_date ASC;
```